



Society of Petroleum Engineers

SPE-230149-MS

From Static to Smart: Elevating Efficiency with Intelligent Valve Technology

H. K. Alrasbi, Occidental Oman

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/230149-MS](https://doi.org/10.2118/230149-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

A control valve is an important device used in industrial systems to control the flow of fluids like gas, steam, water, oil, or chemicals. It works by opening or closing to change the size of the fluid passing through it. This allows it to adjust how much fluid passes through based on instructions from a controller. As a result, it helps keep key conditions such as pressure, temperature, level, and flow at the desired levels.

Control valves work by moving an internal part like a plug or disk, using a power operated actuator. As this internal part moves, it changes the flow area allowing for fluid to pass through.

The actuator provides the force to move the valve stem or plug, and it can be powered pneumatically, electrically, or hydraulically. The positioner ensures the valve reaches the correct opening based on the control signal, this adjustment helps keep process variables as close as possible to their target values.

Introduction

A control valve is an important device used in industrial systems to control the flow of fluids like gas, steam, water, oil, or chemicals. It works by opening or closing to change the size of the fluid passing through it. This allows it to adjust how much fluid passes through based on instructions from a controller. As a result, it helps keep key conditions such as pressure, temperature, level, and flow at the desired levels.

Control valves work by moving an internal part like a plug or disk, using a power operated actuator. As this internal part moves, it changes the flow area allowing for fluid to pass through.

The actuator provides the force to move the valve stem or plug, and it can be powered pneumatically, electrically, or hydraulically. The positioner ensures the valve reaches the correct opening based on the control signal, this adjustment helps keep process variables as close as possible to their target values.

Control valves are divided into two main types based on their movement: linear and rotary. Linear valves, such as globe and diaphragm valves, use straight line motion to control flow, whereas rotary valves, like ball and butterfly valves, regulate flow through a rotating motion. Each type is selected based on the application, required control precision, and fluid characteristics. Factors such as flow rate, pressure drop, temperature, fluid viscosity, and corrosiveness are carefully considered during selection.

With advancements in technology, smart control valves now integrate diagnostic and communication capabilities, allowing predictive maintenance and real-time monitoring. This helps reduce downtime and improves overall plant performance.

Because of their critical role in process control and their ability to react quickly to changes, control valves are widely used in industries such as oil and gas, power generation, chemical processing, and water treatment. The control valves are vital instruments that bridge the gap between control systems and physical processes, ensuring stable and efficient operation across a wide range of industries.

Main parts of a control valve

A control valve consists of several interconnected parts, each serving a specific function to ensure precise regulation of fluid flow within a process system. Understanding these parts is important for designing, operating, and maintaining control valves in industrial systems.

The valve body is the main part of the valve, and it houses the internal components responsible for throttling and is available in various configurations, such as globe, ball, and butterfly designs. The body design influences pressure drop, flow capacity, and compatibility with piping systems.

Within the body lies the valve trim, which includes the plug, seat, stem, and sometimes a cage. The plug modulates the fluid flow by moving relative to the seat, which acts as a sealing surface when the valve is closed. The stem connects the actuator to the plug; it transmits the motion from the actuator to move the plug vertically. In some designs, a cage surrounds the plug, guiding its movement and influencing the valve's flow characteristics and stability. It is important to select the appropriate material to withstand process conditions such as pressure, temperature, and corrosion.

The actuator is responsible for moving the valve's plug by converting control signals into mechanical motion. Common actuator types include pneumatic, electric, and hydraulic, with pneumatic actuators being the most widely used due to their reliability and speed of response.

Attached to the actuator is the positioner, an essential device responsible for ensuring precise valve positioning. It receives the control signal, compares it to the actual valve stem position, and adjusts the actuator output to achieve the desired valve opening. This feedback mechanism enhances precision, particularly in dynamic processes or where valve friction is significant.

The bonnet is the top part of the valve body that covers the valve internals and provides housing for the stem and packing. Bonnet designs vary depending on pressure class and service conditions, including options for extended bonnets used in high or low temperature applications.

Surrounding the stem within the bonnet is the packing, a sealing element designed to prevent process fluid from leaking to the atmosphere. Proper packing selection and maintenance are essential to achieving both leak tight performance and smooth stem movement.

Additional accessories may include handwheels for manual override, limit switches for position feedback, air filter regulators for pneumatic supply, and solenoid valves for on/off signal control. These accessories improve the valve's functionality and allow it to adapt more effectively to specific process needs. Every component within a control valve plays a vital role in ensuring its overall performance, reliability, and precision in control.

A clear understanding of these parts allows engineers and technicians to select, operate, and troubleshoot control valves effectively in a wide range of industrial applications.

Smart control valve

When we refer to smart control valves, we are typically referring to the valves equipped with a smart digital positioner. The intelligence and advanced features come from the positioner, which is mounted on the valve and controls its movement while providing communication and diagnostic capabilities.

Smart control valves are advanced flow control devices designed to enhance automation, monitoring, and performance in modern process systems. Unlike conventional valves, which rely on basic actuator movement controlled by a signal. Smart valves integrate digital intelligence that allows them to communicate, diagnose, and respond in real time. These valves have become increasingly common in industries that prioritize precision, safety, and efficiency.

One of the advantages of smart control valves is that they can often be installed on existing systems without major changes or disruptions. Many industrial plants already have piping, control systems, and field devices in place, and tearing all of that out to upgrade would be costly and time consuming. The good news is that smart valves are designed to work with most current infrastructure. This makes it possible to upgrade gradually and with minimal downtime.

At the core of a smart control valve is a digital positioner that contains built-in sensors and microprocessors. This positioner receives commands from a process controller such as a Distributed Control System (DCS) or a Programmable Logic Controller (PLC) and ensures accurate valve positioning in response to process demands. At the same time, it continuously monitors the valve's behavior, internal conditions, and key process variables like temperature, pressure, and flow rate.

Smart valves are designed to communicate this information through standard digital protocols such as HART, Foundation Fieldbus, or Profibus. These protocols allow the valve to integrate seamlessly with existing control systems without requiring major modifications to wiring or infrastructure. In most cases, an existing conventional valve can be replaced by a smart valve using the same mounting and electrical connections.

Once smart control valves are installed, they offer advanced features like remote configuration, live diagnostics, and continuous performance tracking. They can spot early warning signs such as leaks, wear, or changes in how the valve moves. For example, if the actuator starts to stick or the stem does not move smoothly, the valve sends an alert. This allows operators to perform maintenance based on actual valve condition rather than following a fixed schedule of maintenance. Predictive maintenance not only reduces unplanned downtime but also extends the operational life of equipment.

Smart valves can be calibrated and tuned using handheld communicators or control system software without the need for manual adjustment on site. This feature is especially valuable in hazardous areas or places that are hard to reach where manual intervention poses safety risks. Automatic calibration routines also reduce human error and ensure consistent accuracy across multiple installations. Another important advantage of smart control valves is their ability to record data and track operational history. These valves can log key information such as the percentage of valve opening, the number of strokes, total operating time, and any recent faults.

In the oil and gas sector, regulatory compliance, safety, and equipment reliability are critical due to the high-risk nature of operations and the potential environmental impact of failures. Smart control valves play an important role in supporting these requirements by continuously recording operational data such as valve position changes, stroke counts, response times, and detected faults. This data becomes extremely valuable during safety audits, equipment certification reviews, and incident investigations.

For example, during pipeline operations or refining processes, regulatory bodies may require proof that flow, and pressure were maintained within safe operating limits. The historical data logged by smart valves can serve as evidence that equipment functioned properly throughout a production run or during a specific operational window. In the event of an abnormal shutdown or near-miss, the recorded diagnostics help engineers trace what happened, when it occurred, and whether valve performance contributed to the issue.

Additionally, for critical valves installed in hazardous areas, such as those involved in high pressure flare systems or emergency shutdown lines, smart valve diagnostics can demonstrate whether the valve was responsive and fully operational during testing or actual activation. This level of traceability helps satisfy safety case requirements, ensures compliance with industry standards like API and IEC, and ultimately supports more reliable and transparent operations.

Smart control valves also support two-way communication, allowing operators to issue commands, make adjustments, and receive realtime feedback without physical contact with the valve. Integration with distributed control systems (DCS) or programmable logic controller (PLC), and asset management systems (AMS) enables centralized control and monitoring, improving operational oversight.

Industries such as oil and gas, power generation, water treatment, and chemical manufacturing benefit significantly from smart valve technology. These environments often require high levels of automation, reliability, and real-time monitoring. Smart valves provide the intelligence and control needed to meet those demands while supporting continuous improvement in system performance.

Conventional control valve

On the other hand, conventional control valves in industrial systems are used to regulate the flow of fluids, including gases, liquids, and steam. These valves function by adjusting the position of a movable element in response to a signal from a process controller. Although they do not include digital features like smart valves, conventional control valves are widely used due to their simplicity, reliability, and cost-effectiveness.

In a typical conventional system, the valve gets a control signal either pneumatic or electrical that tells the actuator how much to open or close the valve. The most common signals used are 4-20 mA analog electrical signal (works in one way only) or 3-15 psi pneumatic signal. The actuator then translates this signal into mechanical motion, opening or closing the valve to control flow based on the process requirements.

The main components of a conventional valve include the valve body, actuator, positioner, stem, plug, seat, bonnet, and packing. The valve body is the pressure boundary and flow path, while the actuator provides the force to move the internal trim.

One of the key characteristics of conventional valves is their lack of self-diagnostic or communication capabilities. They do not monitor their own condition or performance, and they do not provide real-time feedback to control systems. Maintenance is typically performed on a fixed schedule or in response to visible performance issues, such as leakage, slow response, or process instability. As a result, unplanned shutdowns and reactive maintenance can sometimes occur, especially in older systems or harsh operating conditions.

Training and maintenance for conventional valves are typically easier for personnel without specialized digital skills. Operators can often troubleshoot and repair issues using basic tools and standard procedures, which is especially beneficial in remote locations or facilities with limited access to technical support. This is why it is still used in some industrial systems today.

Smart vs. Conventional control valves

As the control valves are used across industries to regulate fluid flow and maintain stable process conditions. While both smart and conventional control valves serve this core function, there are significant differences between the two in terms of technology, functionality, operation, and maintenance. Understanding these differences is essential for selecting the right valve type for a specific process or application.

The key difference is in how each type of valve communicates with the control system. Conventional control valves operate based on analog signals, typically 4-20 mA or 3-15 psi, and adjust the valve position mechanically.

These valves do not provide feedback to the control system. Smart control valves, on the other hand, are equipped with digital positioners and sensors that support two-way communications. They not only receive control signals but also send back information about valve status, process conditions, and internal health.

Smart control valves are equipped with advanced diagnostics. They can detect issues such as valve wear, stem friction, seat leakage, and actuator failure before they become critical. This predictive maintenance capability helps prevent unplanned downtime and improves overall system reliability. By continuously monitoring internal performance and health parameters, smart valves can alert operators to developing problems in real time. This allows the maintenance team to act early either during planned inspections or brief process pauses rather than reacting to full equipment failures.

In contrast, conventional valves lack diagnostic features. Maintenance is typically based on fixed schedules or carried out only after noticeable performance problems arise. This reactive approach means that if a valve fails unexpectedly during operation, it can lead to an unplanned shutdown of the process or unit. The consequences of such a shutdown can be significant production losses, increased safety risks,

and delays in identifying the root cause. Without built-in diagnostics, technicians may spend valuable time manually troubleshooting the system to isolate the faulty valve and determine the nature of the issue whether it is air supply failure, a stuck stem, worn internals, or anything else.

These delays extend the total downtime and may even require spare parts or tools that were not initially prepared, adding more time to shut down and complicating recovery efforts. In high throughput facilities, even a few hours of downtime can result in major financial losses, missed delivery targets, or failure to meet customer's demand. By contrast, the early warnings provided by smart control valves support planned interventions, reduce emergency repair work, and help the operation team make faster, more informed decisions ultimately lowering the risk of costly, unexpected shutdowns.

Another key difference between conventional and smart control valves is how they are set up and calibrated. Conventional valves usually need to be adjusted manually, which means a technician must go to the field, connect instruments, and fine-tune the valve by hand. This process can take a lot of time, especially if many valves need to be calibrated. It also opens the door to human errors, small mistakes in settings or alignment can affect how the valve performs later in operation. If something was missed, it might not be noticed until the plant is running, which can lead to delays or performance issues.

On the other hand, smart control valves are designed for much easier and faster setup. Technicians can connect to the valve remotely using a laptop or HART communicator. The system walks them through the configuration process step by step, reducing the chances of mistakes. In many cases, smart valves can even calibrate themselves automatically. For example, the valve can move through a full open and close cycle, measure the limits, and adjust its settings based on real-time feedback all without needing manual input.

This means that smart valves save both time and effort during commissioning and maintenance. Setup is not only quicker, but also more reliable. The built-in tools also help verify that the valve is installed and working correctly before the process starts, which reduces the chances of problems during operation. Over time, this efficiency leads to better performance and fewer issues in the plant.

When it comes to monitoring, smart control valves provide continuous updates on their performance and operating status. These valves can store historical data such as stroke count, opening percentage, and last maintenance activity. This information helps engineers analyze trends, optimize performance, and document operational history. Conventional valves do not store or transmit data, so tracking performance over time is more difficult.

Cost is another area where the two types differ. Conventional valves generally have a lower initial purchase price, making them an attractive option for simpler or stable processes. However, smart valves often offer better long-term value due to reduced maintenance costs, improved process control, and the ability to avoid downtime through early detection of problems. In critical applications, the extra investment in smart valves is usually worthwhile because of the reliability, efficiency, and early problem detection they provide.

Smart control valves are best suited for systems that require a high degree of automation, frequent changes in process conditions, or remote monitoring. They are common in industries such as oil and gas, chemical processing, or power generation. Conventional control valves, while lacking advanced features, remain appropriate for basic applications with steady operating conditions, minimal variation, and limited need for digital integration.

In conclusion, the main differences between smart and conventional control valves involve communication capabilities, diagnostics, configuration, and long-term operational benefits. Smart valves offer enhanced functionality and adaptability for modern automated systems, while conventional valves continue to provide reliable and straightforward control where simplicity and cost-effectiveness are the priority.

Observations

Smart control valves have been in service for many years within oil facilities, particularly in critical process areas where reliability and real-time monitoring are essential. From the early stages of implementation, the use of digital positioners has significantly improved the accuracy and consistency of valve response, especially when compared to older, analog-controlled units. As performance data accumulated and clear improvements were observed in control stability, valve behavior, and maintenance outcomes, confidence in the technology grew steadily. This eventually led to the decision to adopt smart valve technology as a standard across process units.

One of the most noticeable outcomes was the reduction in manual intervention. Previously, field technicians were frequently required to inspect valves showing signs of poor response or erratic behavior. In many cases, the root causes would turn out to be minor issues like stem friction, or air leaks, all of which now get flagged early through internal diagnostics. The alert system integrated into the smart positioners allowed the team to plan intervention before failure occurred, and that alone helped reduce the number of reactive maintenance tasks. Over time, there was a decline in the need to send an operator for onsite troubleshooting, and maintenance shifted toward a more predictive approach. This did not only improve equipment reliability, but also safety, by reducing unnecessary exposure to operating areas.

The team also found that the data logging and self-monitoring features played a major role during performance reviews and troubleshooting. Historical stroke counts, open/close percentages, and fault event logs helped us correlate valve activity with process conditions. It became easier to pinpoint where control issues were originating, whether they were caused by instrument air quality, oversized valves, or even improper tuning in the control system. These insights made process troubleshooting more targeted, and downtime investigations became faster and more accurate.

Smart valve diagnostics also proved valuable during shutdown periods. As part of routine maintenance planning, diagnostic scans are now used to assess valves conditions in advance. Valves with high cycle counts or a history of repeated alerts are flagged for inspection or replacement. This early screening allowed the team to avoid removing valves that were still in good condition and focused only on the ones that needed attention. As a result, the team was able to plan spare parts more efficiently, reduce the overall maintenance workload, and cut unnecessary costs.

The self-calibration and configuration features were another practical advantage, especially during commissioning and instrument loop checks. Field teams were able to perform valve setup faster, with more consistency. The interface was simple to use and easy to navigate, allowing technicians to quickly understand and operate the system. The real-time feedback also enabled quick verification of the correct installation. In the past, valve misalignment or reversed action sometimes went undetected until the process was live, causing delays or needing manual correction. With the smart system in place, those issues were caught early and resolved on the spot.

Looking at long-term performance, we also noticed that process stability improved in loops equipped with smart valves. The valve responded more smoothly, with greater accuracy in positioning. There was less oscillation in control loops, fewer operator interventions, and more time operating within target ranges. Over time, this translated to better overall process efficiency.

In overall, the deployment of smart control valves across operations has proven to be a strong step toward modernization. The combination of real-time diagnostics, digital communication, and improved response accuracy added both operational and maintenance value. From set up to performance monitoring and lifecycle planning, the benefits were clear and measurable. The positive impact extended across departments from maintenance to operations to engineering and reinforced the importance of investing in smart valve technologies as part of our reliability and automation strategy.

Conclusion

Smart control valves represent a major step forward in the field of industrial automation and process control. By combining traditional valve mechanics with digital intelligence, they offer clear advantages in performance, reliability, and operational efficiency. One of their biggest strengths is remote configuration and monitoring. Technicians can access valve diagnostics, adjust settings, or run performance checks without needing to go on-site. This is especially useful in large plants or in hazardous or hard-to-reach areas, where physical access is limited. The ability to detect and report issues early helps reduce emergency repairs, improve planning, and lower overall maintenance costs.

Smart valves also play a critical role in condition-based maintenance. Instead of following fixed maintenance intervals, the system tracks real usage data such as stroke counts, valve response times, and internal wear, allowing maintenance to be performed when needed. This approach not only saves time and resources it also helps prevent early valve replacements or breakdowns caused by late maintenance actions. As a result, the valves last longer and perform more reliably over time.

Importantly, smart control valves can often be installed within existing infrastructure. With support for widely used communication protocols like HART, they can integrate easily into current DCS or PLC systems without needing major upgrades. This makes it feasible for plants to modernize gradually replacing conventional valves as they age and upgrading step by step to a more advanced and cost-effective setup.

While smart valves bring many benefits, conventional control valves still hold a valuable place in industry. They are typically less expensive up front, simpler to maintain, and highly reliable in stable processes where advanced diagnostics are not necessary. For many operations, especially those with consistent process conditions and limited digital infrastructure, conventional valves remain a practical and proven solution.

Acknowledgment

I want to thank my family for always being by my side. Their love, support, and patience gave me the strength to focus and complete this paper. Without them, this journey would not have been possible.

I would also like to thank my company for giving me the chance to write this paper and participate in ADIPEC. Their support and encouragement gave me the opportunity to share my ideas and grow in my professional path. I extend my sincere appreciation to the Ministry of Energy and Minerals for their support in publishing this work.

Finally, I want to mention the role of AI, which helped me check my grammar, and organize my thoughts in a clearer way.

And to everyone who supported me in any way, big or small, I am truly grateful.

References

- Clayton, D. 2018. *What Are Control Valves?* ARCweb Blog. January 1, 2018. What are Control Valves? | ARC Advisory Group
- Don, W. 2022. Control Valves 101: Valve Types, Applications, Components, and Accessories. *THINKTANK*. May 6, 2022. <https://cncontrolvalve.com/control-valves-101-valve-types-applications-components-and-accessories/>
- COVNA, 2025. What is Valve Trim? *COVNA*. May 23, 2025. <https://www.covnav.com/What-is-Valve-Trim-id42526806.html>
- Mahmood, S. Control Valve Working and Types with Parts. Terms Use in Control Valve. *SlideShare Presentation*. <https://www.slideshare.net/slideshow/control-valve-working-and-types-with-parts-terms-use-in-control-valve/148118310#12>
- Mortenson, T. 2020. What Is a Control Valve? *Types, Functions, and Applications*. RealPars Blog. September 7, 2020. <https://www.realpars.com/blog/control-valve>
- SPECS Valve, 2020. The selection of control valve bonnet. *SPECS Valve Knowledge*. March 22, 2020. <https://www.specsvalve.com/The-selection-of-control-valve-bonnet-n.html>
- Vorste, J. 2025. *Instrumentation and control engineering*, first edition. Abu Dhabi, UAE.